

The Large-Scale Neural Systems Simulator Called **iqr**

Looking for a graphical interface to design neuronal models and neural networks? 'iqr' is the perfect free solution for you. This article gives you an idea on how to start working with it

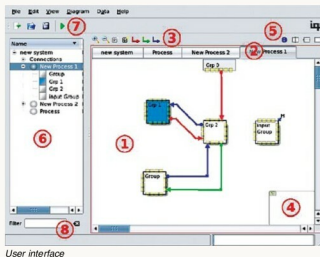
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Neural networks are one of the artificial intelligence techniques that allow you to create 'smart' systems. These networks have also recently created a new frontier in power electronics and other segments within electronics.

iqr, which we will discuss in this article, is a simulation tool for large-scale neural networks. It introduces students to neuronal simulation and neuromorphic control. If you are looking for a graphical interface to design neuronal models, iqr is the perfect free solution. Not only does it provide a graphical interface but also graphical online control of the simulation.

Some of its important features are that we can change the model parameters at run time, and that it has predefined interfaces for robots, cameras and other hardware. It also presents a multi-level neuronal simulation environment to the user. Its modular architecture is open, enabling easy writing of new neurons, synapses and interfaces to new hardware.

Currently, there are alternative simulation tools available that you might already be familiar with, like MATLAB, for instance. However, iqr delivers an advantage over high-level general-purpose simulation tools like MATLAB, since it has predefined building blocks that allow the easy and fast creation of complex systems.



User interface

their details, and also with their ability to interact with the real world.

System requirements

OS: Windows, Mac OS, Linux

HDD space: 34.9 MB

Note. The installation file can be found in the EFY Plus DVD that accompanies this magazine.

TABLE I
Various Command Line Options
that iqr Supports

Command	Function
-f <filename> --file <filename>	Opens the file <filename>
-c <filename> --config <filename>	Opens the configuration file <filename>
-r --run	Starts simulation
-v --version	Displays the version

Also, iqr's approach differs from highly-specific low-level simulation software like Neuron, which focuses more on the thoroughness of the model than its size and the speed of simulation. iqr, on the other hand, is more concerned about connectivity between the neuronal models than

clicking on its icon. To select multiple processes or groups, press the Control key and select. Defining a new process will add a new tab(2) and a new diagram. This Tab bar can be used to switch between processes. The leftmost Tab presents the system level.

Working with iqr

1. Starting up iqr. To start, double click on the IQR icon on the desktop.

The various command line options that iqr supports are listed in Table I.

2. The user interface. Diagram editing pane(1) adds processes, groups and connections to the model. A single process or group is selected by